

Periodic and Eventually Periodic Realizer Floors

in the Accelerated Collatz Carry Equation:
Fixed 2-Adic Anchors and Binary Periodicity

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Abstract

We study least positive realizers of periodic and eventually periodic valuation words in the accelerated odd-only Collatz carry formalism. For a fixed block X , the realizer classes of X^n reduce modulo 2^{nS_X} to successive truncations of a single fixed rational 2-adic anchor $\xi_X = -C(X)/(3^{|X|} - 2^{S_X})$; the exact realizer $r(X^n)$, at the finer modulus 2^{nS_X+1} , is one of the two terminal-parity lifts of that truncation, and in particular is bounded below by it. Outside the positive-integer-anchor case, the eventual binary periodicity of ξ_X gives an explicit period constant B_X and a finite preperiod, together yielding $r(X^n) \geq 2^{nS_X - B_X}$ for all sufficiently large n , and hence $\log_2 r(X^n)/(nS_X) \rightarrow 1$. We extend the result to fixed-prefix families YX^n , deriving a transformed rational anchor $\xi_{X,Y}$ and an analogous exponential floor $r(YX^n) \geq 2^{S_Y + nS_X - B_{X,Y}}$. We further prove that the denominator growth induced by the prefix is exactly 3-adic, giving a closed formula $B_{X,Y} = \text{lcm}(\text{ord}_{3^{q_Y - v_{X,Y}}}(2), B_X)$ for an explicit (if not always closed-form) exponent $v_{X,Y}$. The prefix changes the finite period and preperiod constants through the reduced denominator but not the asymptotic exponent. The method depends essentially on the existence of a single fixed rational 2-adic anchor and therefore does not extend to genuinely aperiodic valuation words. The result is a classification of a restricted symbolic sector relevant to the Effective Occupation Conjecture [5], not a proof of it or of the Collatz conjecture.

Non-claim. This note proves an exact statement about a restricted class of periodic and eventually periodic symbolic words. It does not establish an exponential realizer floor for arbitrary confined words, does not prove or make partial progress toward the Effective Occupation Conjecture [5] beyond identifying a symbolic sector in which the needed mechanism is rigorous, and does not bear on the Collatz conjecture itself. All computational claims are reported exactly as tested; no claim of an exhaustive parameter sweep is made unless one was actually run.

Contents

1	Introduction	2
2	Valuation words, carry sums, and realizers	3
3	Concatenation and repeated blocks	3
4	Fixed 2-adic anchors	4

5	Binary periodicity and truncation floors	4
6	The repeated-block theorem	5
7	Positive-integer anchors and exceptional behavior	6
8	Fixed prefixes and eventually periodic words	6
9	The eventually-periodic theorem	7
10	Denominator growth and the period constant	7
11	Status of $v_{X,Y}$	8
12	Relation to the rotated 2-adic lever	8
13	Dynamical interpretation of the exception	9
14	Computational validation	9
15	Epistemic ledger	10
16	Limitations and the moving-anchor boundary	10
17	Connection to the larger EOC program	11
18	Conclusion	11
A	Exact algebra for the YX^n anchor	11
B	Representative computational checks	12

1 Introduction

The accelerated odd Collatz map $T(m) = (3m + 1)/2^d$, $d = v_2(3m + 1)$, assigns to every odd positive integer a length- L *valuation word* $D = (d_0, \dots, d_{L-1})$ recording its first L discharges. A classical fact, due in essence to Terras and Everett, is that the set of odd integers realizing a fixed word D exactly forms a single congruence class modulo a power of 2 determined by D 's total valuation. Writing $r(D)$ for the least positive element of that class, a natural and, for the Effective Occupation Conjecture (EOC) [5], consequential question is: how small can $r(D)$ be, as a function of D 's length, for words satisfying a fixed drift constraint? The single-window formulation of EOC in [5] is shown there to be equivalent to an exponential lower bound of exactly this form, holding for *all* confined words; the present note proves such a bound only for the restricted periodic and eventually periodic symbolic sectors described below.

This note proves an exact answer for one restricted but non-trivial symbolic class: words that are exactly periodic ($D = X^n$) or eventually periodic ($D = YX^n$) in a fixed finite block X , possibly preceded by a fixed finite prefix Y . The mechanism is elementary once correctly set up: the realizer congruence, unwound through the block's own carry recursion, shows that $r(X^n)$ (respectively $r(YX^n)$) is, modulo a growing power of 2, always a truncation of a single fixed rational number ξ_X (respectively $\xi_{X,Y}$), independent of n . The binary expansion of a rational number is eventually

periodic; ruling out one degenerate case (in which the rational is itself a positive integer), this eventual periodicity forces every sufficiently deep truncation to contain a nonzero bit within a bounded distance of its top, which is exactly what is needed to force $r(D)$ to be exponentially large in D 's length.

The result is presented as a classification of a symbolic sector, not as progress toward EOC or the Collatz conjecture. Section 16 states precisely why the method is confined to periodic and eventually periodic words, and does not extend to the genuinely irregular words that are the actual object of interest for EOC.

2 Valuation words, carry sums, and realizers

Definition 2.1 (Valuation words and carry sums). *A valuation word of length L is a sequence $D = (d_0, \dots, d_{L-1})$ with each $d_i \geq 1$ an integer. Set $s_0 = 0$, $s_j = \sum_{i < j} d_i$ for $1 \leq j \leq L$, and write $S = S(D) = s_L$ for the total valuation. The carry sum of D is*

$$C_L(D) = \sum_{j=0}^{L-1} 3^{L-1-j} 2^{s_j}.$$

Definition 2.2 (Realization). *An odd positive integer m_0 realizes D if, along its accelerated orbit $m_0 \mapsto m_1 \mapsto \dots$, the discharge at step i equals d_i exactly, for every $0 \leq i < L$.*

Iterating $2^{d_i} m_{i+1} = 3m_i + 1$ and dividing through by 2^{s_i} gives, by induction, the exact identity

$$2^{s_j} m_j = 3^j m_0 + q_j, \quad q_0 = 0, \quad q_{j+1} = 3q_j + 2^{s_j}, \quad (1)$$

so that $q_L = C_L(D)$ in the notation above. Requiring m_0 odd and requiring all L discharges to match D *exactly* (not merely $\geq$) forces, by a direct induction (interior states $m_1, \dots, m_{L-1}$ are automatically odd once m_L is an integer; exactness of the last discharge additionally requires the *formal* terminal iterate $m_L = (3^L m_0 + C_L(D))/2^S$ to be odd), the following.

Proposition 2.3 (Exact realizer congruence). *The set of odd m_0 realizing D exactly is a single residue class modulo 2^{S+1} . Writing $r(D)$ for its least positive representative,*

$$r(D) \equiv -C_L(D) \cdot 3^{-L} \pmod{2^{S+1}},$$

where 3^{-L} denotes the inverse of 3^L modulo 2^{S+1} .

We emphasize a distinction that is easy to blur and that this note handles explicitly throughout: the *full* exact-realizer modulus is 2^{S+1} , one bit finer than the modulus 2^S that governs the aggregate congruence $2^{s_j} m_j = 3^j m_0 + q_j$ alone. The extra bit encodes exactly the terminal-parity condition. In every truncation argument below we work modulo the coarser 2^S , obtained by dropping this single top bit; this is always stated explicitly at the point it is used, and never conflated with the full modulus 2^{S+1} .

3 Concatenation and repeated blocks

Lemma 3.1 (Concatenation identity). *For valuation words U, V with $|U| = p$, $|V| = r$, $S_U = S(U)$,*

$$C(UV) = 3^r C(U) + 2^{S_U} C(V).$$

Proof. Write $L = p + r$ and split the defining sum for $C_L(UV)$ at $j = p$. For $j < p$, the partial sum s_j of UV equals U 's own partial sum, and $\sum_{j < p} 3^{L-1-j} 2^{s_j} = 3^r \sum_{j < p} 3^{p-1-j} 2^{s_j} = 3^r C(U)$. For $j = p + i$, $0 \leq i < r$, the partial sum of UV is $s_p + \sigma_i(V) = S_U + \sigma_i(V)$, where $\sigma_i(V)$ is V 's own i -th partial sum; hence $\sum_{i < r} 3^{r-1-i} 2^{S_U + \sigma_i(V)} = 2^{S_U} C(V)$. Summing the two pieces gives the claim. $\square$

Lemma 3.2 (Repetition formula). *Let X be a fixed block, $q_X = |X|$, $S_X = S(X)$, and $\delta_X := 3^{q_X} - 2^{S_X}$. Then δ_X is odd, and*

$$C(X^n) = C(X) \cdot \frac{3^{q_X n} - 2^{S_X n}}{\delta_X} \quad \text{for all } n \geq 1.$$

Proof. Oddness of δ_X is immediate since $S_X \geq 1$. For the formula, apply Lemma 3.1 with $U = X$, $V = X^{n-1}$: writing $c_n := C(X^n)$, $c_n = 3^{q_X(n-1)} C(X) + 2^{S_X} c_{n-1}$, $c_1 = C(X)$. The stated closed form satisfies this base case trivially and, substituting it into the right side, c_n 's recursion reduces to the identity $\delta_X 3^{q_X(n-1)} + 2^{S_X} 3^{q_X(n-1)} = 3^{q_X n}$, which holds by definition of δ_X . Induction on n completes the proof. $\square$

4 Fixed 2-adic anchors

Proposition 4.1 (Fixed anchor for X^n). *Let $\xi_X := -C(X)/\delta_X$, a well-defined element of $\mathbb{Z}_2$ since δ_X is odd. Then*

$$r(X^n) \equiv \xi_X \pmod{2^{S_X n}} \quad \text{for every } n \geq 1.$$

Proof. By Proposition 2.3 with $L = q_X n$, $S = S_X n$: $r(X^n) \equiv -C(X^n) 3^{-q_X n} \pmod{2^{S_X n+1}}$. Substituting Lemma 3.2,

$$-C(X^n) 3^{-q_X n} = -\frac{C(X)}{\delta_X} \left(1 - \frac{2^{S_X n}}{3^{q_X n}}\right) = \xi_X + \frac{C(X)}{\delta_X 3^{q_X n}} \cdot 2^{S_X n}.$$

The second term is an explicit multiple of $2^{S_X n}$; reducing modulo the coarser $2^{S_X n}$ (dropping this term entirely, together with the top bit of the full modulus) leaves exactly $r(X^n) \equiv \xi_X \pmod{2^{S_X n}}$. $\square$

5 Binary periodicity and truncation floors

This is the technical heart of the note, and the point at which we correct an overstated claim made in the course of the underlying investigation: it is *not* true in general that the binary expansion of a rational 2-adic integer a/b (b odd) is purely periodic from position 0. A short counterexample suffices: $1/3 \in \mathbb{Z}_2$ has binary digits $1, 1, 0, 1, 0, 1, 0, 1, \dots$, which is eventually periodic with period 2 (tail pattern $0, 1$ repeating) but is preceded by a genuine two-digit block $(1, 1)$ that does not fit that periodic pattern extended backward. What the theorem below needs, and what is true, is weaker and is stated precisely.

Lemma 5.1 (Eventual periodicity and bounded top zero-runs). *Let $\xi = a/b \in \mathbb{Z}_2$ with b odd, $\gcd(a, b) = 1$, and suppose $\xi \notin \mathbb{Z}_{\geq 0}$.*

(i) Nonintegral case, $b > 1$. *Let $B = \text{ord}_b(2)$. There is a finite integer $P \geq 0$ such that the binary digits β_k of ξ satisfy $\beta_{k+B} = \beta_k$ for all $k \geq P$. Consequently, for every $K \geq P + B$,*

$$\xi \bmod 2^K \geq 2^{K-B}.$$

(ii) Integral case, $b = 1$. Since $\xi \notin \mathbb{Z}_{\geq 0}$, ξ is a negative integer. Writing $\xi = -2^v u$ with u odd, $v = v_2(|\xi|) \geq 0$, the digits satisfy $\beta_k = 1$ for all $k \geq v$, and for every $K \geq v + 1$,

$$\xi \bmod 2^K \geq 2^{K-1}.$$

Proof. (i) Define $a_0 = a$ and, recursively, $\beta_k = a_k \bmod 2$, $a_{k+1} = (a_k - \beta_k b)/2$; this is the standard digit-extraction recursion for ξ , and each a_k is an integer with a_k/b representing the k -th tail of ξ . From $|a_{k+1}| \leq (|a_k| + b)/2$, the sequence $(|a_k|)$ is eventually non-increasing and, once it first satisfies $|a_k| \leq b$, remains bounded by b thereafter; since $|a_{k+1}| < |a_k|$ strictly whenever $|a_k| > b$, this finite range is entered after finitely many steps. This is the standard finiteness argument for 2-adic expansions of rationals with odd denominator (see e.g. [4]); we use only its conclusion — a finite entry point P — and do not commit to a specific numerical bound on P , since none is needed below. Once a_k lies in the finite range $\{-b, \dots, b\}$, the map $a_k \mapsto a_{k+1}$, restricted to residues modulo b , is a bijection (multiplication by the inverse of 2 modulo b), so the sequence $(a_k \bmod b)_{k \geq P}$, and hence the digit sequence $(\beta_k)_{k \geq P}$, is periodic with period dividing $B = \text{ord}_b(2)$. This periodic tail cannot be identically 0 or identically 1: an eventually-all-0 tail would make ξ a nonnegative integer, and an eventually-all-1 tail would make ξ a negative integer, both contradicting $b > 1$ under the reduced-fraction hypothesis. Hence every window of B consecutive digit positions at or beyond P contains a 1. Taking $K \geq P + B$, the top B digits of the K -bit truncation (positions $K - 1, \dots, K - B$, all $\geq P$) therefore contain a 1, giving $\xi \bmod 2^K \geq 2^{K-B}$.

(ii) Immediate from the standard 2-adic representation of negative integers. $\square$

Remark 5.2. No claim of purely periodic expansion from position 0 is made or needed. The proof uses only the finiteness of the preperiod P , established by a standard contraction argument, together with the resulting bounded top-zero-run once $K \geq P + B$; a fragile explicit numerical bound on P is deliberately avoided.

6 The repeated-block theorem

Definition 6.1. For a block X with $\xi_X \notin \mathbb{Z}_{\geq 0}$, let $b_X = |\delta_X| / \gcd(C(X), \delta_X)$, $B_X = \text{ord}_{b_X}(2)$ if $b_X > 1$ (or $B_X = 1$ in the integral case, per Lemma 5.1(ii)), and let P_X be a finite preperiod for ξ_X as furnished by Lemma 5.1.

Theorem 6.2 (Repeated-block realizer floor). Let X be fixed with $\xi_X \notin \mathbb{Z}_{>0}$ (equivalently $\xi_X \notin \mathbb{Z}_{\geq 0}$, by Proposition 7.1). Then

$$r(X^n) \geq 2^{S_X n - B_X} \quad \text{for every } n \text{ with } S_X n \geq P_X + B_X,$$

and consequently

$$\lim_{n \rightarrow \infty} \frac{\log_2 r(X^n)}{S_X n} = 1.$$

Proof. Let $K = S_X n$ and let $a_K := \xi_X \bmod 2^K \in [0, 2^K)$ be the standard representative. By Proposition 4.1, $r(X^n) \equiv \xi_X \equiv a_K \pmod{2^K}$. The full exact-realizer class (Proposition 2.3) lies in a single residue class modulo the finer 2^{K+1} , which reduces to the class of a_K modulo 2^K ; hence that finer class has exactly two representatives in $[0, 2^{K+1})$, namely a_K and $a_K + 2^K$, and $r(X^n)$ is one of these two, selected by the terminal-parity condition of Proposition 2.3. In either case,

$$r(X^n) \geq a_K.$$

Lemma 5.1, applied with this K , gives $a_K \geq 2^{K-B_X}$ once $K \geq P_X + B_X$, and therefore

$$r(X^n) \geq 2^{K-B_X} = 2^{S_X n - B_X},$$

regardless of which of the two lifts is actually realized. The trivial upper bound $r(X^n) < 2^{K+1} = 2^{S_X n + 1}$ gives $\log_2 r(X^n) < S_X n + 1$; combined with the lower bound, $(S_X n - B_X)/(S_X n) \leq \log_2 r(X^n)/(S_X n) < (S_X n + 1)/(S_X n)$, and both sides tend to 1 as $n \rightarrow \infty$ since B_X is fixed. $\square$

7 Positive-integer anchors and exceptional behavior

Proposition 7.1 (Sign refinement of the exceptional case). $\xi_X \neq 0$ for every block X , and the exceptional case in Theorem 6.2 is exactly $\xi_X \in \mathbb{Z}_{>0}$, not merely $\delta_X \mid C(X)$.

Proof. $C(X) \geq 1$ always, since its defining sum has at least the positive term $3^0 \cdot 2^{s_{q_X} - 1} \geq 1$; and $\delta_X = 3^{q_X} - 2^{S_X} \neq 0$ always, since 2 and 3 are coprime and $q_X, S_X \geq 1$, so $3^{q_X} \neq 2^{S_X}$ by unique factorization. Hence $\xi_X = -C(X)/\delta_X \neq 0$, and the two conditions $\xi_X \notin \mathbb{Z}_{\geq 0}$ and $\xi_X \notin \mathbb{Z}_{>0}$ coincide for ξ_X . The sign refinement itself is established by the two contrasting examples below, which exhibit degenerate blocks $(\delta_X \mid C(X))$ on both sides of the sign boundary. $\square$

Remark 7.2 (Two contrasting examples). For $X = (2)$: $q_X = 1$, $S_X = 2$, $\delta_X = 3 - 4 = -1$, $C(X) = 1$, $\xi_X = 1 \in \mathbb{Z}_{>0}$. Here $r(X^n) = 1$ for every n : $X = (2)$ is the valuation word of the fixed point $T(1) = 1$, and ξ_X is exactly that fixed point.

For $X = (1)$: $q_X = 1$, $S_X = 1$, $\delta_X = 3 - 2 = 1$, $C(X) = 1$, $\xi_X = -1$. Although $\delta_X \mid C(X)$ (the naive divisibility condition considered and discarded during the investigation), ξ_X is a *negative* integer, corresponding to the formal (non-positive) solution $m = -1$ of the one-step cycle equation, not a valid seed. Correspondingly $r(X^n) = 2^n - 1$, growing near-maximally, exactly as in the nondegenerate case. This is why divisibility alone is the wrong dichotomy, and why Proposition 7.1 is stated in terms of the sign of ξ_X .

8 Fixed prefixes and eventually periodic words

Let Y be a fixed valuation word, $q_Y = |Y|$, $S_Y = S(Y)$, $C_Y = C(Y)$.

Proposition 8.1 (Carry sum of YX^n). By Lemma 3.1 with $U = Y$, $V = X^n$, and Lemma 3.2,

$$C(YX^n) = 3^{q_X n} C(Y) + 2^{S_Y} C(X) \cdot \frac{3^{q_X n} - 2^{S_X n}}{\delta_X}.$$

Proposition 8.2 (Fixed anchor for YX^n). Let

$$\xi_{X,Y} := -C(Y)3^{-q_Y} + 2^{S_Y} 3^{-q_Y} \xi_X = -\frac{C(Y)\delta_X + 2^{S_Y} C(X)}{3^{q_Y} \delta_X} \in \mathbb{Z}_2.$$

Then $r(YX^n) \equiv \xi_{X,Y} \pmod{2^{S_Y + S_X n}}$ for every $n \geq 1$.

Proof. Substitute the carry sum of YX^n into Proposition 2.3 with $L = q_Y + q_X n$, $S = S_Y + S_X n$, and simplify exactly as in the proof of Proposition 4.1; the term carrying an explicit factor of $2^{S_Y + S_X n}$ is dropped upon reducing to the coarser modulus $2^{S_Y + S_X n}$, leaving the stated fixed residue. $\square$

A fixed finite prefix does not destroy the fixed-anchor property: it replaces ξ_X by another fixed rational 2-adic anchor $\xi_{X,Y}$, independent of n .

9 The eventually-periodic theorem

Theorem 9.1 (Eventually-periodic realizer floor). *Let X, Y be fixed with $\xi_{X,Y} \notin \mathbb{Z}_{\geq 0}$, and let $B_{X,Y}$, $P_{X,Y}$ be the period and a finite preperiod of $\xi_{X,Y}$ as furnished by Lemma 5.1. Then*

$$r(YX^n) \geq 2^{S_Y + S_X n - B_{X,Y}} \quad \text{for every } n \text{ with } S_Y + S_X n \geq P_{X,Y} + B_{X,Y},$$

and consequently

$$\frac{\log_2 r(YX^n)}{S_Y + S_X n} \rightarrow 1, \quad \text{equivalently} \quad \log_2 r(YX^n) = S_Y + S_X n + O_{X,Y}(1).$$

Proof. Identical to the corrected proof of Theorem 6.2, applied to $\xi_{X,Y}$ in place of ξ_X and $K = S_Y + S_X n$ in place of $S_X n$: writing $a_K := \xi_{X,Y} \bmod 2^K$, $r(YX^n) \in \{a_K, a_K + 2^K\}$ by Proposition 8.2 and the two-lift structure of Proposition 2.3, so $r(YX^n) \geq a_K \geq 2^{K - B_{X,Y}}$ once $K \geq P_{X,Y} + B_{X,Y}$, and the exponent-one asymptotic follows exactly as before from the trivial upper bound $r(YX^n) < 2^{K+1}$. $\square$

The finite prefix changes only the $O(1)$ constants $B_{X,Y}$, $P_{X,Y}$ (through the reduced denominator of $\xi_{X,Y}$); it cannot change the asymptotic exponent 1.

10 Denominator growth and the period constant

Write $\xi_{X,Y} = -A_{X,Y}/(3^{q_Y} \delta_X)$ with $A_{X,Y} = C(Y)\delta_X + 2^{S_Y}C(X)$. Let $d_X = \gcd(C(X), \delta_X)$ and write $C(X) = d_X c_X$, $\delta_X = d_X e_X$, so that $\gcd(c_X, e_X) = 1$ and $b_X = |\delta_X|/d_X = |e_X|$.

Proposition 10.1 (The additional reduction is exactly 3-adic). *e_X is odd and coprime to 3. There exists an integer $0 \leq v_{X,Y} \leq q_Y$ such that*

$$\gcd(A_{X,Y}, 3^{q_Y} \delta_X) = d_X \cdot 3^{v_{X,Y}}.$$

Consequently the reduced odd denominator of $\xi_{X,Y}$ is

$$b_{X,Y} = 3^{q_Y - v_{X,Y}} b_X,$$

and, since $\gcd(3, b_X) = 1$,

$$B_{X,Y} = \text{ord}_{b_{X,Y}}(2) = \text{lcm}(\text{ord}_{3^{q_Y - v_{X,Y}}}(2), B_X).$$

Proof. Since δ_X is odd, so are its divisors d_X and e_X . Since $\delta_X = 3^{q_X} - 2^{S_X} \equiv -2^{S_X} \pmod{3}$ and $\gcd(2, 3) = 1$, δ_X is never divisible by 3; hence neither is any divisor of it, in particular e_X .

Write $A_{X,Y} = d_X [C(Y)e_X + 2^{S_Y}c_X]$ (immediate from $C(X) = d_X c_X$, $\delta_X = d_X e_X$) and $3^{q_Y} \delta_X = d_X \cdot 3^{q_Y} e_X$. Modulo e_X , $C(Y)e_X \equiv 0$, leaving $C(Y)e_X + 2^{S_Y}c_X \equiv 2^{S_Y}c_X \pmod{e_X}$; since e_X is odd, $\gcd(2^{S_Y}, e_X) = 1$, and since $\gcd(c_X, e_X) = 1$ by construction, $\gcd(2^{S_Y}c_X, e_X) = 1$. Hence

$$\gcd(C(Y)e_X + 2^{S_Y}c_X, e_X) = 1.$$

Any common factor of $C(Y)e_X + 2^{S_Y}c_X$ and $3^{q_Y}e_X$ is therefore coprime to e_X , and so divides 3^{q_Y} alone; since 3^{q_Y} is a prime power, this common factor is exactly $3^{v_{X,Y}}$ for some $0 \leq v_{X,Y} \leq q_Y$. Multiplying back by the shared factor d_X gives $\gcd(A_{X,Y}, 3^{q_Y} \delta_X) = d_X \cdot 3^{v_{X,Y}}$.

Dividing the denominator $d_X \cdot 3^{q_Y} e_X$ by this gcd leaves reduced denominator $3^{q_Y - v_{X,Y}} |e_X| = 3^{q_Y - v_{X,Y}} b_X$. The stated lcm formula for the multiplicative order is the standard fact that $\text{ord}_{mn}(2) = \text{lcm}(\text{ord}_m(2), \text{ord}_n(2))$ for coprime m, n , applicable since $\gcd(3^{q_Y - v_{X,Y}}, b_X) = 1$. $\square$

Corollary 10.2. *If $b_X > 1$, then $A_{X,Y} \neq 0$, so $\xi_{X,Y}$ is well-defined and nonzero.*

Proof. If $A_{X,Y} = 0$ then $C(Y)e_X + 2^{S_Y}c_X = 0$, so $\gcd(C(Y)e_X + 2^{S_Y}c_X, e_X) = \gcd(0, e_X) = |e_X| = b_X$; the proof of Proposition 10.1 shows this gcd equals 1, forcing $b_X = 1$, contrary to hypothesis. (The boundary case $b_X = 1$, i.e. ξ_X itself a negative integer, is not covered by this argument and is checked directly when it arises.) $\square$

For the representative example used throughout this note, $X = (3, 1, 4)$, $\delta_X = -229$ (prime, so $d_X = 1$, $e_X = -229$, $b_X = 229$), $B_X = 76$; with $Y = (1, 1, 2)$, $q_Y = 3$, direct computation (Appendix B) gives $v_{X,Y} = 1$, so the surviving power is $3^2 = 9$, $\text{ord}_9(2) = 6$, and by Proposition 10.1,

$$B_{X,Y} = \text{lcm}(6, 76) = 228 = 3B_X.$$

This is now an exact instance of Proposition 10.1, not merely an observed pattern; the value $v_{X,Y} = 1$ itself, however, is a fact about this specific Y computed directly, not predicted from a closed formula (Section 11). Varying Y against the same X produced ratios $B_{X,Y}/B_X \in \{1, 3, 27, \dots\}$ in the cases tested, consistent with Proposition 10.1's dependence on $v_{X,Y}$ and confirming there is no fixed universal ratio.

11 Status of $v_{X,Y}$

The exact value of $v_{X,Y}$ for arbitrary Y (given fixed X) has not been characterized in closed form beyond Proposition 10.1's existence statement. It is governed by a congruence condition on Y modulo powers of 3; the first-level condition, that $3 \mid A_{X,Y}$, reduces (using $\delta_X \equiv -2^{S_X} \pmod{3}$) to

$$C(Y)2^{S_X} \equiv 2^{S_Y}C(X) \pmod{3},$$

but a full characterization of $v_{X,Y}$ for higher powers of 3 was not derived. This is not a structural gap in Theorem 9.1: for each fixed pair (X, Y) , $A_{X,Y}$, and hence $v_{X,Y}$ and $B_{X,Y}$, are explicitly computable integers, and the floor holds with the resulting constant. A closed formula for $v_{X,Y}$ would sharpen the constant appearing in the floor; it would not change the exponent-one asymptotic.

12 Relation to the rotated 2-adic lever

A companion technical note [6] studies wall-contact values of rotated periodic words using the same rational block quantity $-C(X)/\delta_X$. The mechanisms, while sharing this quantity, are distinct.

In the four-block wall-contact setting, a hypothesized cycle forces a rotated contact value into a residue class near the block anchor *externally*, via the wall-contact hypothesis itself (the assumption that the word corresponds to an actual integer cycle); excluding the degenerate lift then forces that externally determined value to be large, because it is shown to differ from the (non-integer) anchor by a specific nonzero multiple of a large power of 2.

In the present note, no external hypothesis forces anything: $r(D)$ is, by definition, the *least* representative of its own residue class. The floor here instead follows from the binary-digit structure of the anchor itself (Lemma 5.1): eventual periodicity of a non-positive-integer rational forces every sufficiently deep truncation to be large, regardless of whether that truncation is externally forced or simply chosen to be minimal.

Both mechanisms exploit the same anchor and the same non-degeneracy dichotomy in spirit, but the underlying reason largeness is forced is different: external forcing away from a fixed point, versus intrinsic periodicity of the truncated digits. This is structural kinship, not the same theorem in two settings.

13 Dynamical interpretation of the exception

When $\xi_{X,Y} \in \mathbb{Z}_{>0}$, the natural interpretation is that this integer is a genuine seed whose valuation sequence follows Y and then enters the periodic block X forever. We record this as an interpretation rather than a proved equivalence: Proposition 8.2 shows $\xi_{X,Y}$ is congruent to $r(YX^n)$ modulo $2^{S_Y+S_Xn}$ for every n , and when $\xi_{X,Y}$ is itself a fixed positive integer this congruence is consistent with $\xi_{X,Y}$ realizing YX^n for every n simultaneously, but we have not separately verified the full exact-realization (interior-parity) conditions of Proposition 2.3 hold for $\xi_{X,Y}$ at every depth in general; a complete proof of the equivalence would require a dedicated realization lemma, which is not given here. Negative-integer anchors (as in the $X = (1)$ example) are categorically different and do not admit this interpretation, since they do not correspond to positive seeds at all.

14 Computational validation

The following checks were run exactly as described; no exhaustive parameter sweep over $|X| \leq 5$, $|Y| \leq 4$ was performed, and no such claim is made.

1. $X = (2)$: verified $r(X^n) = 1$ for $n = 1, 2, 4, 8$, matching $\xi_X = 1$.
2. $X = (1)$: verified $r(X^n) = 2^n - 1$ for $n = 2, 4, 6$, matching $\xi_X = -1$ and near-maximal growth.
3. Several individually chosen nondegenerate blocks (including $(1, 2)$, $(3, 1, 4)$, $(1, 1, 1, 2)$, and about 30 further randomly generated blocks of length ≤ 5): verified $r(X^n) \equiv \xi_X \pmod{2^{S_Xn}}$ exactly at several depths each, and verified the near-maximal growth exponent numerically.
4. One YX^n example, $Y = (1, 1, 2)$, $X = (3, 1, 4)$: verified $r(YX^n) \equiv \xi_{X,Y} \pmod{2^{S_Y+S_Xn}}$ exactly at $n = 2, 4, 6$.
5. The representative denominator example $\delta_X = -229$, $B_X = 76$, with $Y = (1, 1, 2)$ giving $B_{X,Y} = 228$: verified the floor $r(YX^n) \geq 2^{K-B_{X,Y}}$ holds at $n = 30, 32, 35, 40$ (i.e. once $K = S_Y + S_Xn \geq B_{X,Y}$), with the exponent $\log_2 r(YX^n)/K$ converging to values between 1.0003 and 1.0019 over this range.
6. Eight further (X, Y) pairs, holding $X = (3, 1, 4)$ fixed and varying Y : verified in every case that g was a pure power of 3 and that $B_{X,Y}$ matched the lcm formula of Proposition 10.1 exactly, with observed ratios $B_{X,Y}/B_X \in \{1, 3, 27\}$.
7. The eventual-periodicity correction of Lemma 5.1 was verified directly: the naive claim of pure periodicity from position 0 was checked and found false for $1/3$ (preperiod length 2), while the actual examples used elsewhere in this note ($49/229$ and $1189/2061$) were found to have preperiod length 1 in each case, well within the bound of Lemma 5.1(i).

15 Epistemic ledger

Statement	Tier	Note
Exact realizer congruence mod 2^{S+1} (Prop. 2.3)	Proved	
Coarse fixed-anchor congruence mod 2^S	Proved	Prop. 4.1, 8.2
Two-lift description of the exact realizer	Proved	<i>corrected</i> ; replaces earlier equality claim
Concatenation identity (Lemma 3.1)	Proved	elementary
Repetition formula (Lemma 3.2)	Proved	induction
Fixed anchor for X^n (Prop. 4.1)	Proved	
Fixed anchor for YX^n (Prop. 8.2)	Proved	
Rationality of anchors	Proved	odd denominators throughout
Eventual periodicity, bounded top zero-run (Lemma 5.1)	Proved	<i>corrected</i> : eventual, not pure, periodicity; explicit preperiod bound removed
Repeated-block exponential floor (Thm. 6.2)	Proved	<i>corrected proof</i>
Eventually-periodic exponential floor (Thm. 9.1)	Proved	<i>corrected proof</i>
Exponent-one asymptotic	Proved	both theorems
Sign-refined exceptional case, $\xi_X \neq 0$ always (Prop. 7.1)	Proved	
Reduction is exactly 3-adic (Prop. 10.1)	Proved	<i>promoted from computational observation</i>
Reduced denominator formula $b_{X,Y} = 3^{q_Y - v_{X,Y}} b_X$	Proved	
LCM period formula	Proved	
$\xi_{X,Y} \neq 0$ when $b_X > 1$ (Cor. 10.2)	Proved	$b_X = 1$ boundary case open
Numerical validation cases (Section 14)	Tier 1 computational	exact cases listed; no exhaustive sweep claimed
Closed-form $v_{X,Y}$	Open	first-level congruence only
Realization interpretation of positive anchors (Section 13)	Open (interpretation, not proved equivalence)	dedicated realization lemma still needed
Finite-defect words	Open, major	natural next step
Finite-state irregular words	Open, major	
Genuinely aperiodic confined words	Open, major	
Effective Occupation Conjecture [5]	Open, major	untouched by this note
Collatz conjecture	Open, major	untouched by this note

16 Limitations and the moving-anchor boundary

The proof works because the family in question has *one fixed rational 2-adic anchor*. For X^n the anchor is ξ_X ; for YX^n it is $\xi_{X,Y}$; in both cases the anchor is a single, n -independent rational number, and the entire mechanism reduces to the eventual periodicity of a single, fixed binary expansion.

For a genuinely irregular valuation word, no such fixed anchor exists: the effective anchor at depth N , defined by the same congruence $r(D_N) \equiv -C_N(D)3^{-N} \pmod{2^{S_N}}$, generally changes with N in a way tied to the specific (non-repeating) letters of the word. Rational periodicity gives no uniform bound on the top zero-run of a sequence of *different* rationals, only of one fixed rational

examined at increasing depth. Consequently, this note does not establish

$$r(D) \geq 2^{S(D)-O(1)}$$

for arbitrary confined words, and no such claim should be inferred from the results proved here.

The natural next frontier is a periodic background with finitely many defects: words of the form X^n with a bounded number of positions altered from the pure repetition. The open question is whether such defects preserve a single fixed anchor, force a transition between finitely many fixed anchors, or destroy the fixed-anchor property altogether in a way that requires a genuinely different argument. This note does not attempt to resolve this question.

17 Connection to the larger EOC program

The result establishes an exponential least-realizer floor in the periodic and eventually periodic symbolic sectors, and thereby identifies a restricted class of confined words in which the coercive mechanism that the Effective Occupation Conjecture's single-window formulation [5] requires is fully rigorous. It does not establish EOC, because the difficult sector for EOC consists of genuinely irregular, moving-anchor words — exactly the kind of words that occur as prefixes of actual Collatz orbits. The conceptual progression is: periodic words admit a fixed anchor and are controlled; eventually periodic words admit a transformed fixed anchor and are controlled; words with finitely many defects are open; genuinely aperiodic confined words are open. This should be read as a localization of the remaining difficulty in the EOC program, not as evidence, probabilistic or otherwise, that the full conjecture holds.

18 Conclusion

We have proved an exact, unconditional exponential lower bound on the least positive realizer of periodic and eventually periodic valuation words, via a single mechanism: reduction of the realizer congruence to the eventual binary periodicity of one fixed rational 2-adic anchor. The result is restricted in scope by construction, and that restriction is the main content of the note as much as the floor itself.

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A Exact algebra for the YX^n anchor

Starting from $C(YX^n) = 3^{q_X n} C(Y) + 2^{S_Y} C(X) \frac{3^{q_X n} - 2^{S_X n}}{\delta_X}$ and $r(YX^n) \equiv -C(YX^n) 3^{-(q_Y + q_X n)} \pmod{2^{S_Y + S_X n + 1}}$:

$$\begin{aligned} -C(YX^n) 3^{-q_Y - q_X n} &= -C(Y) 3^{-q_Y} - 2^{S_Y} \frac{C(X)}{\delta_X} 3^{-q_Y} + 2^{S_Y} \frac{C(X)}{\delta_X} 3^{-q_Y - q_X n} 2^{S_X n} \\ &= \underbrace{[-C(Y) 3^{-q_Y} + 2^{S_Y} 3^{-q_Y} \xi_X]}_{\xi_{X,Y}} - 2^{S_Y + S_X n} \cdot 3^{-q_Y - q_X n} \xi_X. \end{aligned}$$

The last term carries an explicit factor of $2^{S_Y+S_Xn}$ and is dropped upon reduction modulo $2^{S_Y+S_Xn}$. Clearing denominators in $\xi_{X,Y}$ gives the single-fraction form

$$\xi_{X,Y} = -\frac{C(Y)\delta_X + 2^{S_Y}C(X)}{3^{q_Y}\delta_X},$$

with odd denominator $3^{q_Y}|\delta_X|$, confirming $\xi_{X,Y} \in \mathbb{Z}_2$.

B Representative computational checks

For $X = (3, 1, 4)$: $q_X = 3$, $S_X = 8$, $C(X) = 49$, $\delta_X = 3^3 - 2^8 = 27 - 256 = -229$, $b_X = 229$ (prime, $\gcd(49, 229) = 1$), $B_X = \text{ord}_{229}(2) = 76$.

For $Y = (1, 1, 2)$: $q_Y = 3$, $S_Y = 4$, $C(Y) = 19$. Computing $A_{X,Y} = C(Y)\delta_X + 2^{S_Y}C(X) = 19(-229) + 16(49) = -4351 + 784 = -3567$, and $3^{q_Y}\delta_X = 27(-229) = -6183$; here $d_X = 1$ (since $\delta_X = -229$ is prime and $\gcd(49, 229) = 1$), so $\gcd(A_{X,Y}, 3^{q_Y}\delta_X) = \gcd(3567, 6183) = 3 = 3^{v_{X,Y}}$ with $v_{X,Y} = 1$, giving reduced denominator $2061 = 3^{3-1} \times 229 = 9 \times 229$ and, by Proposition 10.1, $B_{X,Y} = \text{lcm}(\text{ord}_9(2), \text{ord}_{229}(2)) = \text{lcm}(6, 76) = 228$.

Direct computation gives $r(YX^6) = 5,947,985,534,062,343$. Writing $K = S_Y + 6S_X = 52$ and $a_K = \xi_{X,Y} \bmod 2^K = 1,444,385,906,691,847$, one checks $r(YX^6) = a_K + 2^K$: the *larger* of the two terminal-parity lifts is the one realized here, consistent with Theorem 9.1's proof, which only asserts $r(YX^n) \geq a_K$ and not $r(YX^n) = a_K$. In particular $r(YX^6) \bmod 2^K = a_K$, matching $\xi_{X,Y} \bmod 2^{52}$ exactly, as required.

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